

Name: _____

Due: September 9, 11:59PM via e-mail or Edwards Mailbox

1. Which of the following reactions are possible without non-standard model physics? For invalid reactions, indicate what the issue.
 - a. $^{69}\text{Kr} \rightarrow ^{69}\text{Br} + e^+ + \nu_e$
 - b. $^{44}\text{Ti}(\alpha, p)^{48}\text{Cr}$
 - c. $e^- + ^{55}\text{Sc} \rightarrow ^{55}\text{Ca} + \bar{\nu}_e$
 - d. $^{15}\text{O}(\alpha, \gamma)^{19}\text{Ne}$

2. Calculate the Q-values for the following and compare to the NNDC Q-value calculator. Show your work.
 - a. ^{182}Hf β^- decay
 - b. $^{100}\text{Mo}(t, p)$ ($t = \text{triton}$)
 - c. $^{96}\text{Zr}(\alpha, p)$
 - d. $^{54}\text{Fe}(^3\text{He}, n)$

3. Can ^3He and ^3H spontaneously β -decay? Discuss your results.

4. There is a stable isotope of each element within $Z=1-82$, except Pm and what other element?

5. Where is the proton dripline for the chlorine isotopes? (You can check the NNDC chart) Prove it using proton-separation energy calculations.

6. Considering the terms of the 5-parameter liquid drop model, which parameter is responsible for the following feature of the nuclear landscape? Discuss your logic.
 - a. Location of valley of stability for low A?
 - b. Bend of valley of stability away from $N=Z$ for large A?
 - c. Large A/Z ratios?
 - d. Lack of ultra-high A nuclides?
 - e. Existence of nuclides at all?

7. A table of experimental binding energy per nucleon as compiled in the 2012 Atomic Mass valuation has been provided. Fit these data using the 5-parameter liquid drop model and report fit-parameter values. Report the fit function to give the parameters context. Attach to this homework a plot of the fit residuals as a function of neutron number and, separately, as a function of proton number, as well as a copy of the code used to perform the fit (e.g. ROOT, gnuplot, Mathematica).